

How Accurate is the Popular Lumped Orthotropic Thermal Conductivity Approach Employed for PCB Modelling?

In this article, we will discuss about the different PCB (Printed Circuit Board) modelling approaches that are widely used in thermal simulation analysis and also examine their accuracy for the temperature predictions.

Overview of PCB Thermal Modelling Approaches

PCB is a stack-up structure of electrically conducting and insulating layers. Each of the conducting layers is designed with a specific artwork pattern of electrical connections on that conductive layer and the layers are electrically interconnected through the plated holes described as vias.

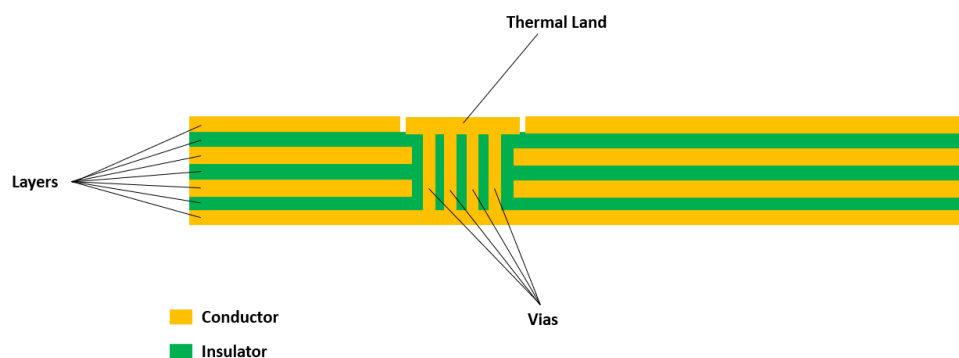
Generally constructing thermal model geometrically exact to the actual PCB with all its elements is challenging, laborious and computational expensive.

Hence the alternative PCB modelling approaches are chosen by the analysts, and the most common approaches are given below:

- Detailed PCB Model
- Isotropic Conductivity PCB Model
- Orthotropic Conductivity PCB Model

Detailed PCB Model

In Detailed PCB Modelling, all conducting and insulating layers are individually modelled as lumped cuboids, and the effective thermal conductivity is assigned to each layer cuboid. Other features like the vias, thermal lands, local copper concentration etc. are discretely modelled.



Detailed PCB Model

The effective thermal conductivity of each layer may be calculated using the volume averaged-method formula as given below.

Effective Thermal Conductivity, $K_e = K_c \times \alpha + K_i \times (1 - \alpha)$

Where

K_c : Thermal Conductivity of Conductor

K_i : Thermal Conductivity of Insulator

α : Fraction of Conductor Coverage of the Layer

The above PCB modelling approach is usually seen as consuming more computational space due to the inclusion of thin geometrical feature i.e., the individual layer thickness, so more geometrically simplified models are preferred by the most analysts.

Isotropic Conductivity PCB Model

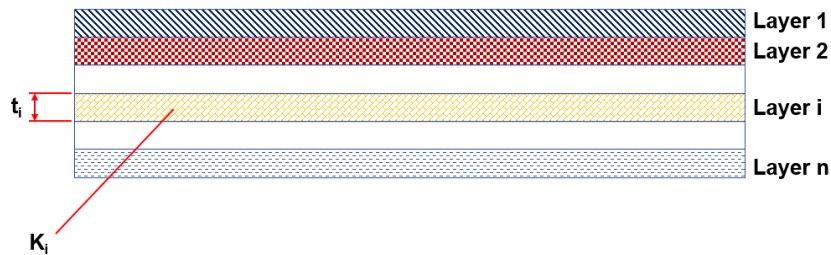
In this approach, the PCB is modelled as a single lumped cuboid with an equivalent thermal conductivity assigned to it. The effective conductivity described in the Detailed PCB Model is applied for the whole PCB cuboid here.

Though this model is quick to develop & solves faster in CFD environment, it finds a limited usage only even during the preliminary analysis phase and mostly not used for the detailed analysis as the accuracy is questionable.

Orthotropic Conductivity PCB Model

This is the very popular approach for PCB modelling, it is same as of the single lumped cuboid method, but the orthotropic thermal conductivities termed as “In-Plane Thermal Conductivity” & “Through-Plane Thermal Conductivity” are assigned to it.

PCB Thermal Conductivity in In-Plane & Through-Plane directions are calculated using the below formulae:



PCB Layer Structure

$$K_{In-Plane} = \frac{\sum_{i=1}^n K_i t_i}{\sum_{i=1}^n t_i}$$

$$K_{Through-Plane} = \frac{\sum_{i=1}^n t_i}{\sum_{i=1}^n \frac{t_i}{K_i}}$$

Where

t_i : Thickness of the i^{th} layer

n : Total no of layers

K_i : Effective conductivity of the i^{th} layer

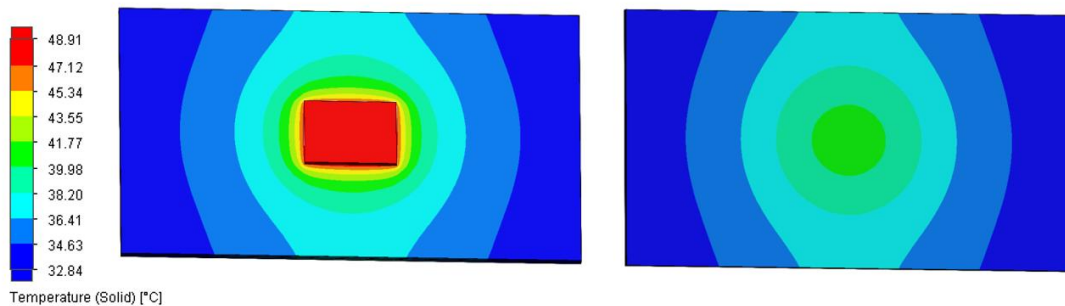
As this approach is widely used, this article attempts to find out the accuracy & limitations of this modelling through the CFD simulation analysis by taking a simple simulation example described below.

Simulation Case

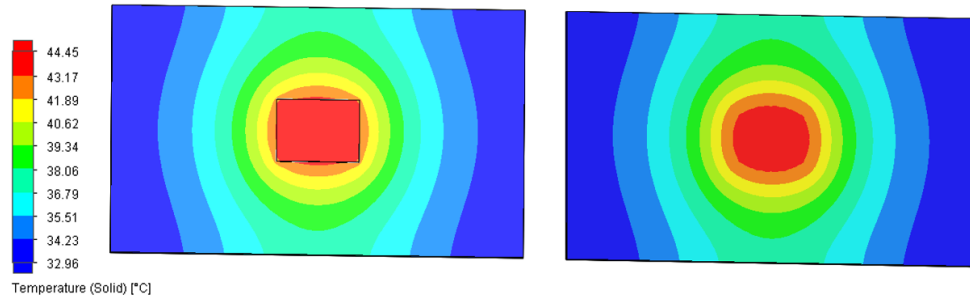
The simulation case considered for the PCB modelling investigation is described below:

- Two signal & two plane (2S2P) layered PCB of size 100.0 x 60.0 x 1.6 mm.
- Each signal is 70 μ m thickness with 30% copper coverage and each plane layer is 35 μ m thickness with 80% copper coverage and dielectric material is taken as FR4.
- Natural cooling method is used in this conjugate heat transfer analysis to attain an active participation of the PCB.
- 2.0 Watts heat load and 20°C ambient conditions are applied in this case.
- Size of the heat source is varied to study its impact on the temperature prediction.

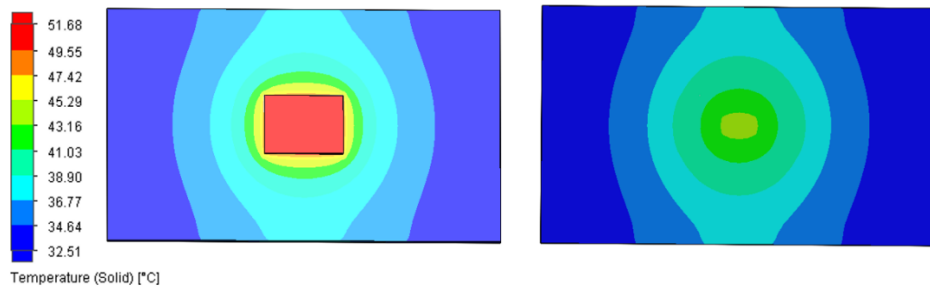
This case helps to understand fairly the significance of different PCB modelling approaches and the simulation results of temperature contours for all three modelling approaches are shown below.



Detailed PCB Model (Both the Sides)



Isotropic Thermal Conductivity PCB Model (Both the Sides)

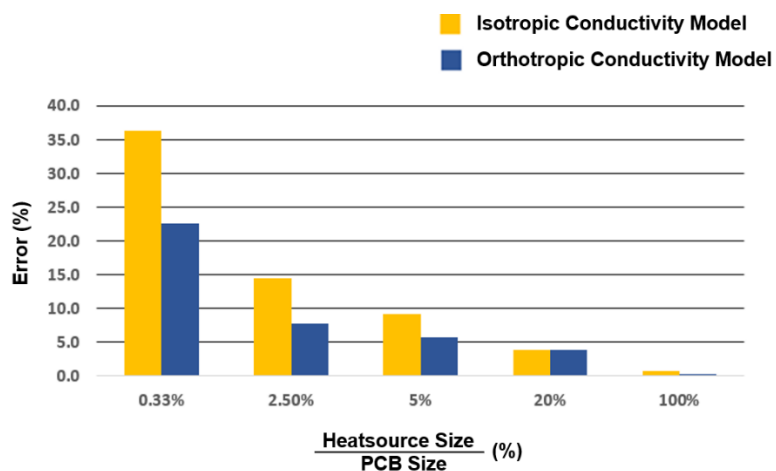


Orthotropic Thermal Conductivity PCB Model (Both the Sides)

Modelling Approach	Component Temperature (Max)	Error (Against the Detailed PCB Model)	Simulation Results Remarks
Detailed PCB	48.9 °C		
Isotropic Conductivity PCB	44.4 °C	-9.2%	Temperature contour is not agreeing well with the detailed PCB Model
Orthotropic Conductivity PCB	51.7 °C	5.7%	Temperature contour is nearly agreeing with the detailed PCB Model

Impact of Heat Source Size on Temperature Prediction

Heatsource size was varied to study the impact on the temperature prediction, and simulation results for this parametric study are given in the below chart.



Impact of Heatsource Size on Temperature Prediction

It can be observed from the above simulation results that the size of heatsource has a major impact on the PCB thermal modelling approach.

The orthotropic conductivity model showed a better result than the isotropic conductivity model for all the simulated cases and the error difference between these models becomes negligible when the heatsource size is nearly close to the PCB size.

It is very important to note that for the small sized heatsource 5.0 x 4.0 mm in our case, the error was found to be very high for both the lumped models.

Conclusion

The level of accuracy & limitation of the popular orthotropic PCB modelling approach can be well understood from the simulation case investigated in this article.

The level of accuracy for this model can further be improved by discreetly modelling the top/bottom copper layer that is contacting the heatsource/component including the local copper concentration, thermal lands, thermal vias etc. appropriately which could further improve the accuracy of the temperature prediction.

References

1. Vinod Nirale, "Evaluation of Effective Thermal Conductivity in PCB", IJIRST – International Journal for Innovative Research in Science & Technology| Volume 3 | Issue 01 | June 2016
2. Rafael Lopes Costa and Valeri Vlassov, "Simulation Method for Effective Thermal Conductivity Determination of Complex Boards", Space Mechanics and Control Division – DMC, National Institute for Space Research.

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